

HR Management System (HRMS)

Database Architecture & Implementation

Team 11 • German International University (GIU)

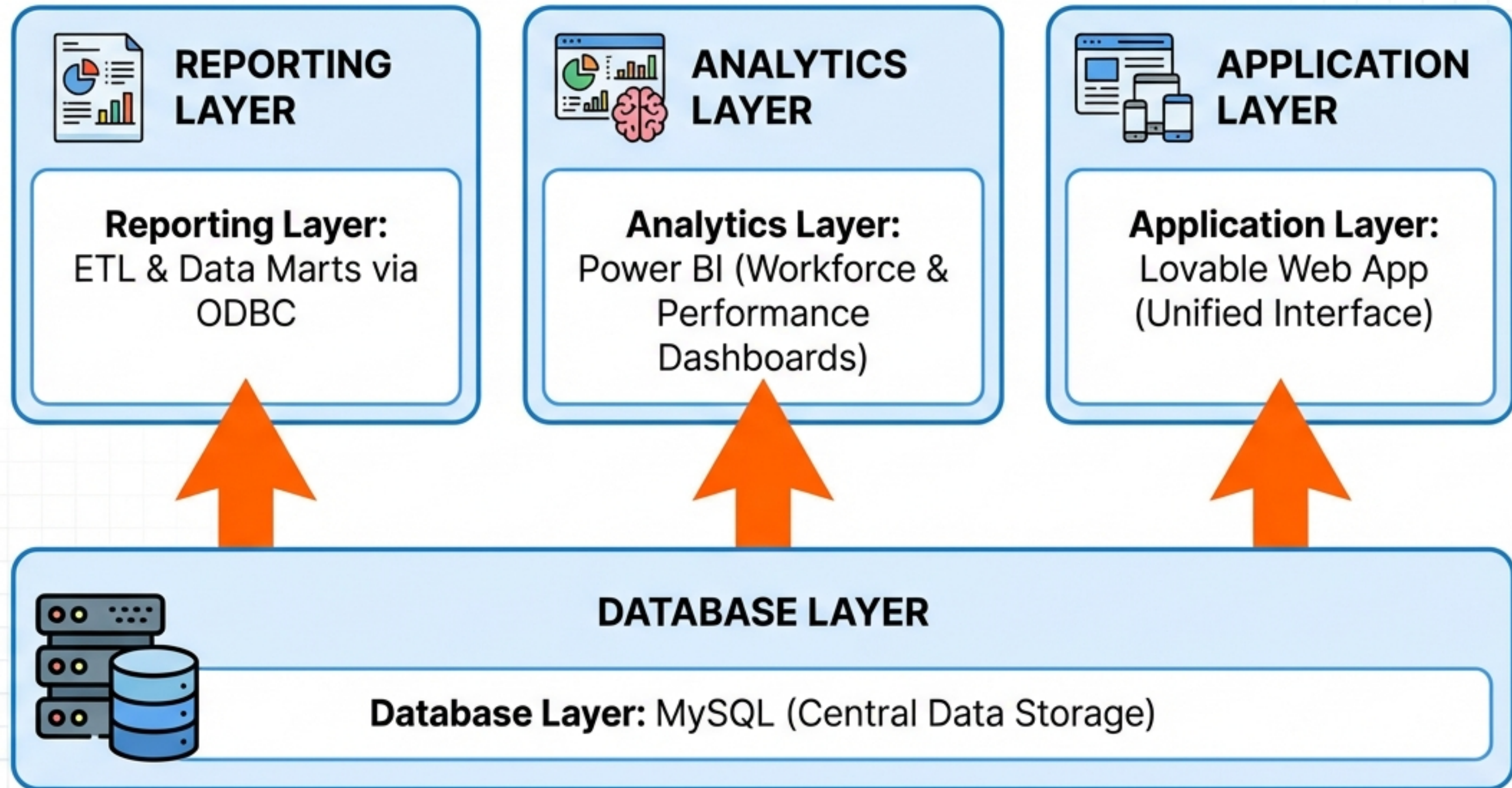
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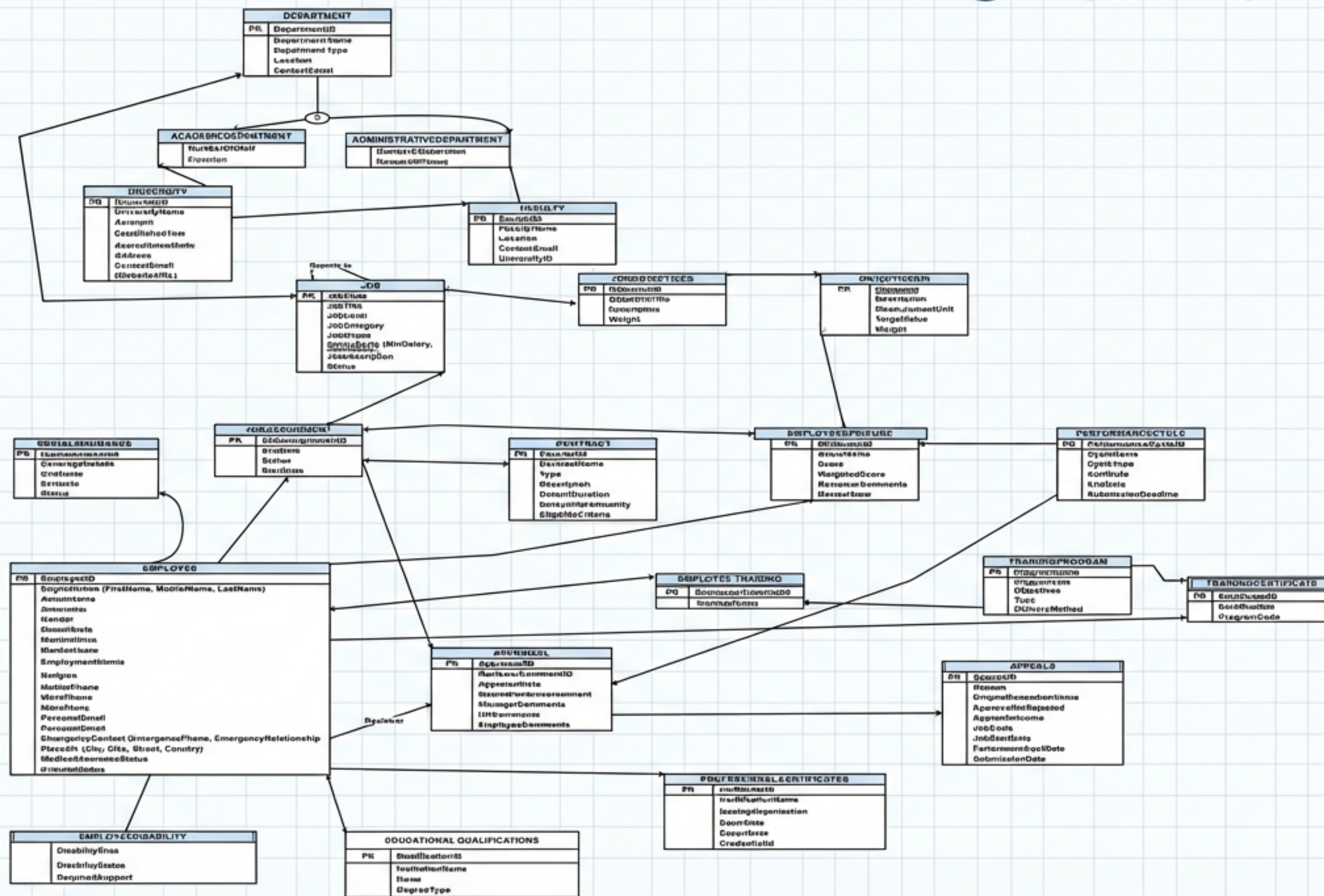
System Architecture & Tech Stack



Development Methodology



Database Design (ERD)



Core Entities:
Employee, Job,
Department,
Performance Cycle,
Appraisal

Schema Scope: 22 Structured Tables

Relationships:
Comprehensive mapping of assignments, training, and KPI scoring.

Implementation Highlights



Schema:

22 Structured
Data Tables

11 Reporting
Views



Logic:

15 Functions

18 Stored
Procedures

6 Triggers



Connectivity:
Views imported
to Power BI via
ODBC



Automation:
Rules for data
consistency

Automated Data Integrity (Triggers)

- **trg_validate_objective_weight:**
Ensures total weights $\leq 100\%$.
- **trg_prevent_delete_objective_with_kpi:** Protects dependencies.
- **trg_prevent_delete_employee_with_active_assignment:** Prevents accidental data loss.
- **trg_auto_calc_weighted_score:** Instantly updates KPI scores.
- **trg_validate_training_certificate:**
Verifies training prerequisites.



Operational Logic (Stored Procedures)

Employee Management

AddNewEmployee

UpdateEmployeeContactInfo

GetEmployeeFullProfile

Job & Assignment

AddNewJob

AssignJobToEmployee

CloseJobAssignment

Performance

CreatePerformanceCycle

AddEmployeeKPIScore

CreateAppraisal

SubmitAppeal

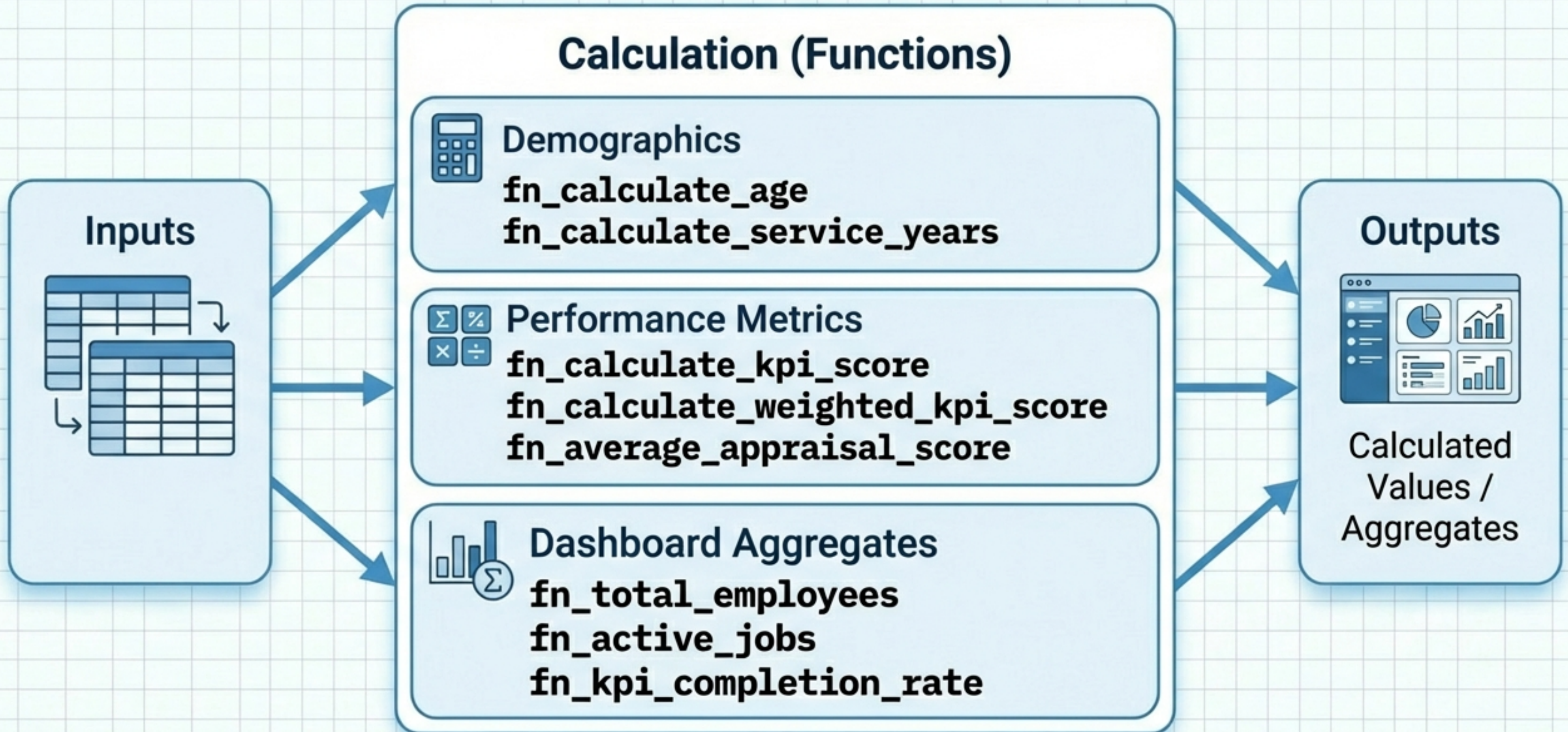
Training

AddTrainingProgram

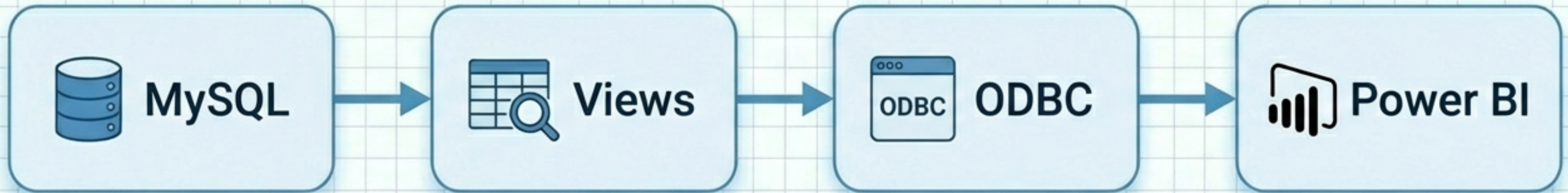
AssignTrainingToEmployee

IssueTrainingCertificate

Computational Logic (Functions)



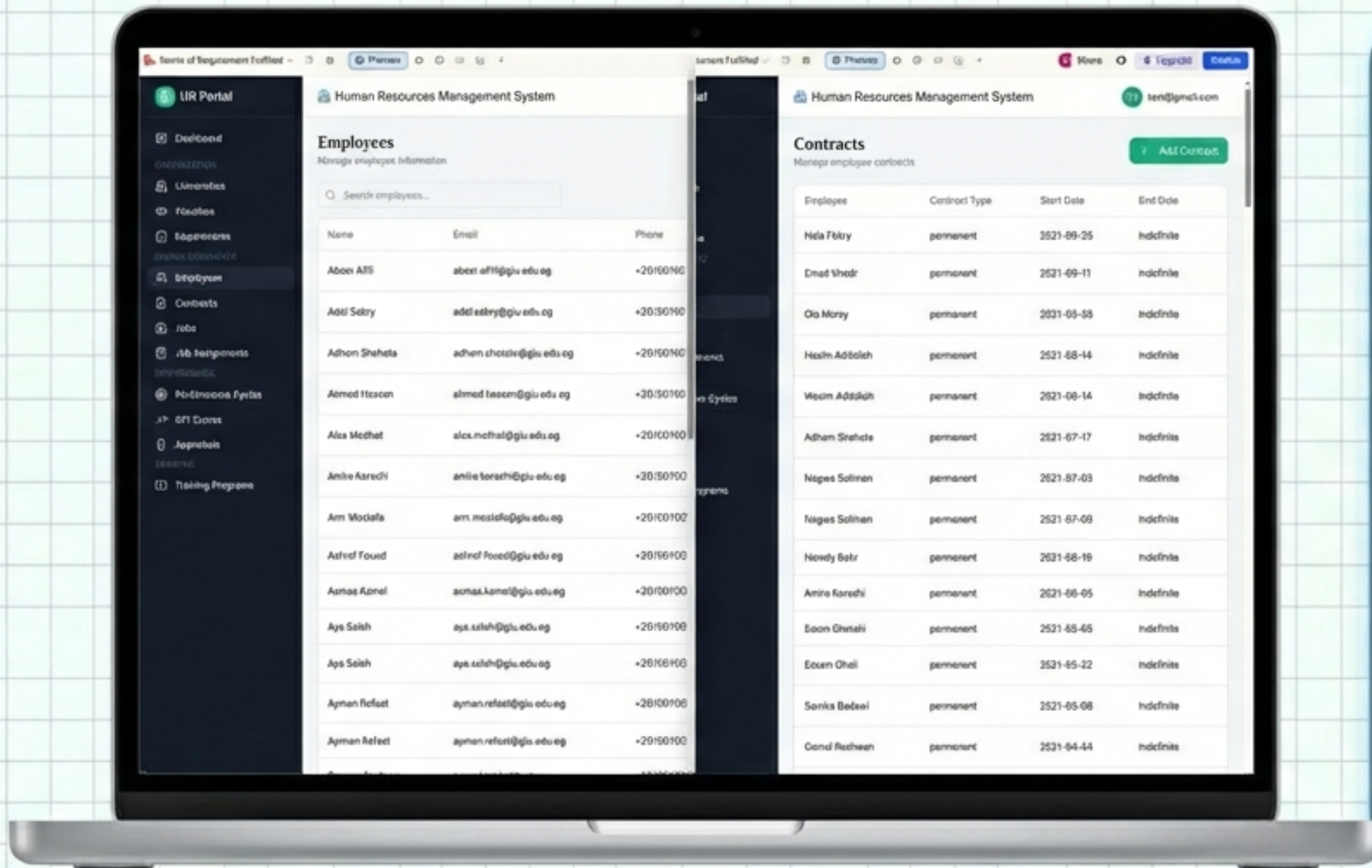
Reporting Views Architecture



Specific Views Created

- **Workforce:** `vw_employee_count_by_department`, `vw_gender_distribution`
- **Job Insights:** `vw_jobs_by_level`, `vw_salary_stats_by_category`
- **Performance:** `vw_kpi_scores_summary`, `vw_appraisal_scores_per_cycle`
- **Training:** `vw_training_completion_stats`

The Web Application



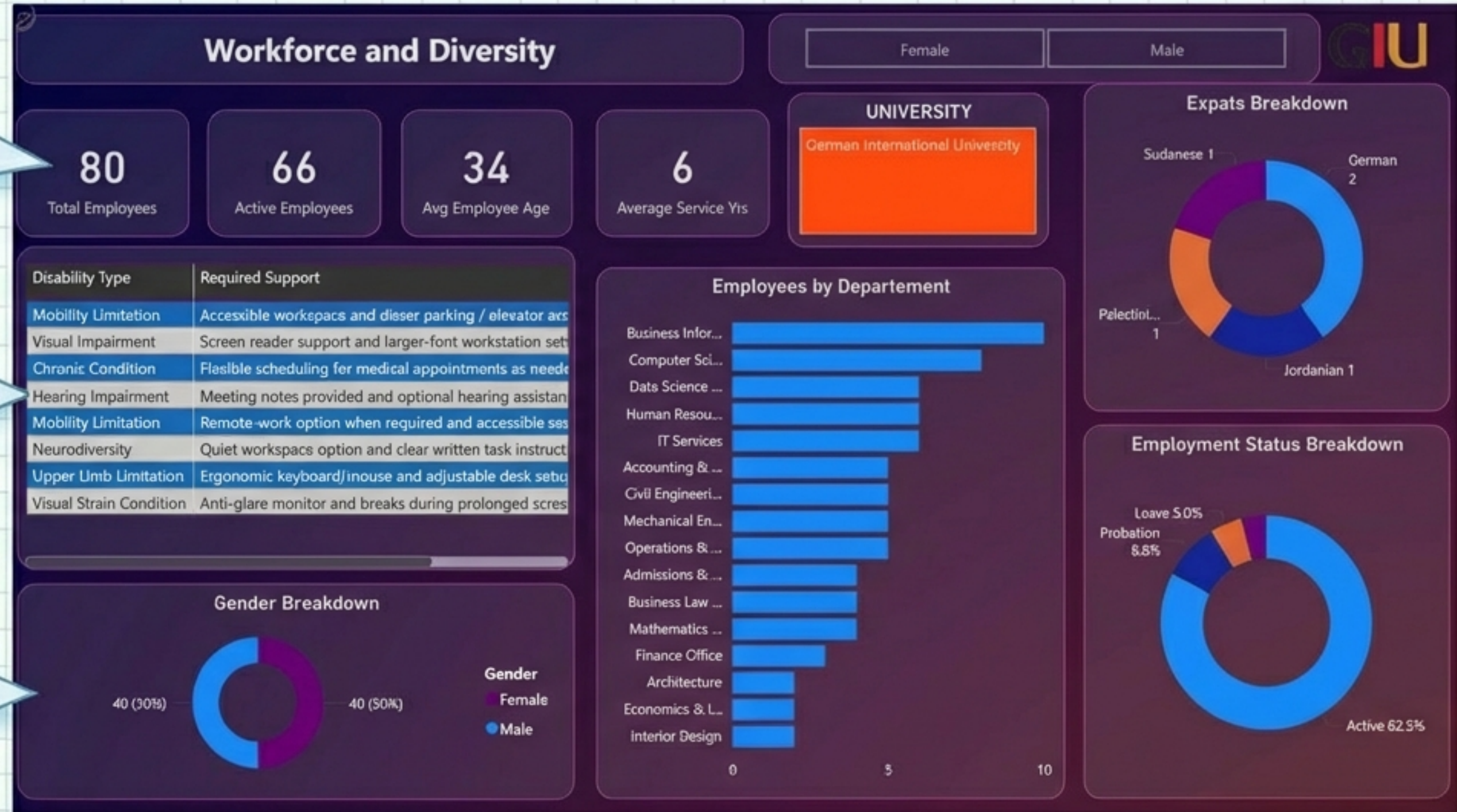
- Built on **Lovable** platform
- Full CRUD capabilities (Create, Read, Update, Delete)
- Modules: Employee Profiles, Contract Management, Digital Appraisals

Analytics: Workforce & Diversity

80 Total Employees,
66 Active

Disability
Support
Tracking

50/50
Gender Split



Analytics: Job Structure & Training



- **Job Structure:** Salary statistics by category (Academic vs Admin).
- **Training:** Participation rates and completion stats per program.
- **Performance:** Appraisal scores trend analysis.

Analytics: Organizational Hierarchy



- **Hierarchical View:** University > Faculties > Departments
- **Detail Level:** Job Codes, Grades, and Salary Bands
- **Position Status:** Frozen vs. Open tracking

Key Outcomes



Unified View of
Workforce & HR Activities



Improved Consistency
of HR Metrics



Enhanced Visibility into
Performance & Training



Faster Access to
Insights for Decisions

Conclusion & Access



<https://databasehrproject.lovable.app>

A fully integrated, automated, and data-driven HR ecosystem.

Q&A